School of Computer Science and Artificial Intelligence

WEEK-11 Assignment # 21

Program : B. Tech (CSE)

Specialization :AIML

Course Title : Competitive Programming

Course Code : 23CS302PC305

Semester : VI

Academic Session : 2025-2026

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Enrollment No. : 2303A52126

Batch No. : 33

Date :31/03/26

Primality Testing

Context

You are developing a financial analytics system that tracks daily transaction counts. The system needs to check whether the total transactions over a specific range of days form a prime number.

Task

- Write a program that supports the following operation:
- CheckPrime(left, right): Compute the sum of transactions from day left to day right (inclusive) and:
- Print "Prime" if the sum is a prime number
- Print "Not Prime" if the sum is not a prime number

Requirements

Input:

7

12 5 7 10 15 6 8

3

CheckPrime 1 3

CheckPrime 2 5

CheckPrime 4 7

Output:

Not Prime

Not Prime

Prime

Explanation:

- Query 1: Sum of days 1–3 → 12+5+7 = 24 → Not Prime
- Query 2: Sum of days 2–5 → 5+7+10+15 = 37 → Prime
- Query 3: Sum of days 4–7 → 10+15+6+8 = 39 → Not Prime

Python:

Assignment21 > Main.py > ...

```python
import math

# Function to check prime
def is_prime(n):
    if n <= 1:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False

    for i in range(3, int(math.sqrt(n)) + 1, 2):
        if n % i == 0:
            return False
    return True


# Input
n = int(input())
arr = list(map(int, input().split()))

# Prefix sum
prefix = [0] * (n + 1)
for i in range(1, n + 1):
    prefix[i] = prefix[i - 1] + arr[i - 1]

# Queries
q = int(input())

for _ in range(q):
    query = input().split()
```

Ln 40, Col 27 Spaces: 4 UTF-8 CRLF { } Python 3.11.9

```python

# Queries
q = int(input())

for _ in range(q):
    query = input().split()
    left = int(query[1])
    right = int(query[2])

    range_sum = prefix[right] - prefix[left - 1]

    if is_prime(range_sum):
        print("Prime")
    else:
        print("Not Prime")
```

Ln 40, Col 27 Spaces: 4 UTF-8 CRLF { } Python 3.11.9

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Assignment21/Main.py"
7
12 5 7 10 15 6 8
3
12 5 7 10 15 6 8
3
3
CheckPrime 1 3
CheckPrime 1 3
Not Prime
Not Prime
CheckPrime 2 5
Prime
CheckPrime 4 7
Prime
CheckPrime 4 7
Not Prime
```

Java:

Main.java > Language Support for Java(TM) by Red Hat > Main

```java
import java.util.*;

public class Main {

    // Function to check prime
    static boolean isPrime(int n) {
        if (n <= 1) return false;
        if (n == 2) return true;
        if (n % 2 == 0) return false;

        for (int i = 3; i * i <= n; i += 2) {
            if (n % i == 0) return false;
        }
        return true;
    }

    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // Prefix sum
        int[] prefix = new int[n + 1];
        for (int i = 1; i <= n; i++) {
            prefix[i] = prefix[i - 1] + arr[i - 1];
```

```java
        int[] prefix = new int[n + 1];
        for (int i = 1; i <= n; i++) {
            prefix[i] = prefix[i - 1] + arr[i - 1];
        }

        int q = sc.nextInt();
        sc.nextLine(); // consume newline

        for (int i = 0; i < q; i++) {
            String line = sc.nextLine();
            String[] parts = line.split(regex: " ");

            int left = Integer.parseInt(parts[1]);
            int right = Integer.parseInt(parts[2]);

            int rangeSum = prefix[right] - prefix[left - 1];

            if (isPrime(rangeSum)) {
                System.out.println(x: "Prime");
            } else {
                System.out.println(x: "Not Prime");
            }
        }

        sc.close();
    }
}
```

```
eptionMessages' '-cp' 'C:\Users\sowmy\AppData\Roaming\Code\User\workspaceSt
4523a538cc2dbf9\redhat.java\jdt_ws\Week11_d710793f\bin' 'Main'
7
34 5 6 3 2 1 7
3
CheckPrime 1 3
Not Prime
CheckPrime 4 6
Not Prime
CheckPrime 2 5
Not Prime
```